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Improvement of the post-processing algorithm in AMR-NB speech codec

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Abstract

Narrow band adaptive multi-rate (AMR-NB) is a speech codec that supports multiple bit-rate modes and is widely used in mobile communication systems. However, the synthesized speech quality of its lower bit-rate modes is not very satisfying. This paper aims to improve the synthesized speech quality of the lower bit-rate modes of AMR-NB (e.g., 4.75kb/s) by optimizing its post-processing algorithm. The optimization mainly focuses on four modules including the formant filter with adjusted correction factor, the fixed codebook gain smoothing, the adaptive gain control and the anti-sparse processing. The experimental results show that the AMR-NB codec with the optimized post-processing algorithm achieves an obvious improvement of the synthesized speech quality, and the lower bit-rate mode achieves the bigger improvement.

Keywords: *Speech codec; AMR; post-processing*

1. Introduction

As the most important communication way in the human communication, speech communication plays a vital role in mobile communication systems. And higher performance is required for speech coding technology with the development of mobile communication. Narrow band adaptive multi-rate (AMR-NB) is a widely used speech coding technology in mobile communication networks. It is a speech codec standard based on code excited linear prediction (CELP) algorithm developed by 3GPP in 1999. It includes 8 kinds of bit-rate modes among

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4.75 ~ 12.2Kb/s and a DTX mode of 1.8Kb/s [1]. However, the synthesized speech quality of its lower bit-rate modes is not very satisfying, which is partly due to its post-processing algorithm.

In fact, post-processing techniques are very important for a speech codec. In recent years, some post-processing algorithms based on signal statistics and environmental characteristics have been proposed to improve the speech quality and intelligibility [2]. For example, by increasing the signal-to-noise ratio of the non-stationary part of the speech signal under the smoothing constraint, the intelligibility of synthesized speech is improved in [3]. The fixed codebook gain smoothing technique can reduce the energy fluctuation of the excitation signal, thereby improving the perceived quality of speech [4]. When the anti-sparse processing algorithm is applied to the fixed codebook vector, the low bit-rate perceptual quality can also be improved [5]. Besides, advanced post-processing algorithm is adopted in enhanced voice services (EVS) codec in order to reduce the noise and interference of audio signals before and after encoding [6].

Therefore, we optimize the AMR-NB post-processing algorithm, namely the formant filter, the adaptive gain control, the fixed codebook gain smoothing and the anti-sparse processing to improve the synthesized speech quality of AMR-NB in the lower bit-rate modes.

The remainder of this paper is organized as follows. In Section 2, the post-processing algorithm of AMR-NB is summarized. Section 3 describes the improved post-processing algorithm. Experiment results and analysis are given in Section 4 which is followed by conclusions.

2. The post-processing algorithm of AMR-NB

The post-processing algorithm of AMR-NB include formant filter, adaptive gain control, fixed codebook gain smoothing and anti-sparse processing. The formant filter is defined as

$$H_f(z) = \frac{\hat{A}(z/\gamma_n)}{\hat{A}(z/\gamma_d)} \tag{1}$$

where the factors γ_n and γ_d control the formant filtering amount.

To compensate for the spectral tilt of $H_f(z)$, a filter $H_t(z)$, is used defined as

$$H_t(z) = 1 - \mu z^{-1} \tag{2}$$

where μ is tilt factor. Also, the AMR-NB achieves adaptive gain control by defining a gain compensation factor γ_{sc}.

$$\gamma_{sc} = \left(\frac{\sum_{n=0}^{39} \hat{s}^2(n)}{\sum_{n=0}^{39} s_f^2(n)} \right)^{1/2} \tag{3}$$

Then the signal after gain compensation is as following.

$$\hat{s}'(n) = \beta_{sc}(n)\hat{s}_f(n) \tag{4}$$

where $\beta_{sc}(n)$ is updated for each sample, and the rule is

$$\beta_{sc}(n) = \alpha\beta_{sc}(n-1) + (1-\alpha)\gamma_{sc} \tag{5}$$

where $\alpha = 0.9$.

In addition, the fixed codebook gain smoothing module is based on a measure of the short-term spectral stationarity in the q-domain. And the average value $\overline{q}$ for frame n is obtained by

$$\overline{q}(n) = 0.84 \cdot \overline{q}(n-1) + 0.16 \cdot \overline{q}_4(n) \tag{6}$$

For each sub-frame, the difference measure between the average vector and the quantized interpolation vector is calculated by the following formula.

$$diff_m = \sum_j \sum_m \frac{\left|\overline{q}^{(j)}(n) - \overline{q}_m^{(j)}(n)\right|}{\overline{q}^{(j)}(n)} \tag{7}$$

The anti-sparseness processing consists of circular convolution of the fixed codebook vector with an impulse response. Three pre-stored impulse responses are used and a number $impNr = 0,1,2$ is set to select one of them. The selection of the impulse response is performed adaptively from the adaptive and fixed codebook gains.

3. The improved post-processing algorithm

By optimizing the AMR-NB post-processing algorithm, namely the formant filter, the adaptive gain control, the fixed codebook gain smoothing and the anti-sparse processing, the synthesized speech quality in the lower bit-rate coding modes can be improved. Fig.1 shows the decoder structure of AMR-NB, where the blue blocks indicate the post-processing modules to be optimized. The optimization scheme of each module will be described in details as following.

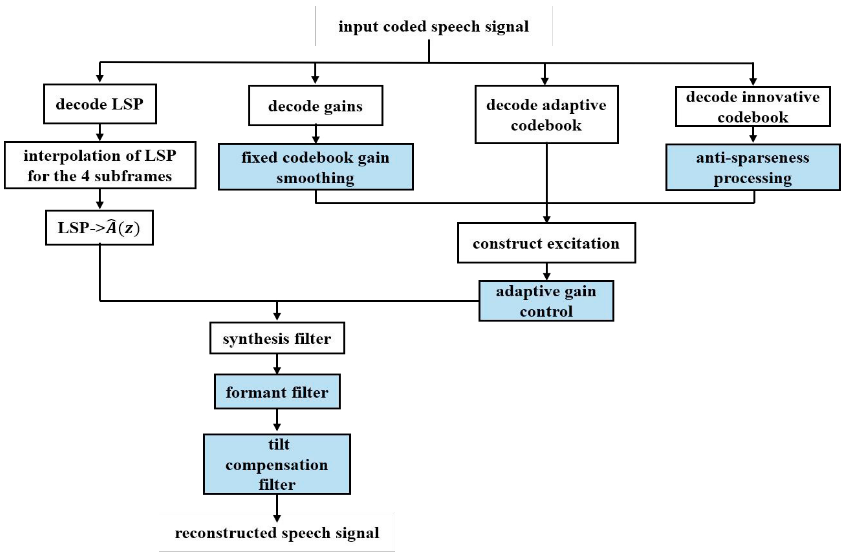

Fig. 1. Decoder of AMR-NB with the improved post-processing algorithm.

3.1. Improved formant filter

The statistical properties of the formant filter are introduced. An adjustment factor g_u is defined as

$$g_u = \sum_{n=0}^{21} \left|h_u(n)\right| \tag{8}$$

where $h_u(n)$ is the Impulse response of $A(z/\beta)/A(z/\alpha)$. Meanwhile, α is the bandwidth expansion factor and β is the tilt compensation factor.

Hence the improved formant filter is defined as

$$H_u(z)=\frac{\hat{A}(\frac{z}{\beta})}{g_u\cdot\hat{A}(\frac{z}{\beta})} \tag{9}$$

By defining the compensation factor $g_t=1-|\gamma_t k_1'|$ to compensate for the decreasing effect of g_u in $H_u(z)$. Then the tilt compensation filter $H_t(z)$ is:

$$H_t(z)=\frac{1}{g_t}(1-\mu z^{-1}) \tag{10}$$

Through the adjustment of the parameters α,μ,g_u and g_t, the cooperation of the formant filter and the tilt compensation become more coordinated, thereby improve the synthesized speech quality.

3.2. Improved adaptive gain control

The improved adaptive gain control can allow the energy of the speech signal before and after filtering to be approximately the same.

Since each speech frame vector produces only one gain adjustment factor. If no low-pass filter is added, the gain adjustment factor will have a staircase effect. Low-pass filter can effectively smooth this step effect, and generally use point-by-point correction to eliminate discontinuities between signal samples.

$$g(n)=\alpha g(n-1)+(1-\alpha)G, n=0,...,L, g(-1)=1.0 \tag{11}$$

The coefficient α affects the low-pass filtering effect. The larger the value of α, the narrower the bandwidth of the low-pass filter, and the smoother the gain adjustment factor. However, when α is larger, the instantaneous gain adjustment effect cannot be achieved, and rapid changes in amplitude cannot be easily tracked. When α is smaller, the low-pass filtering effect will be worse, which is not conducive to eliminating the step effect and the discontinuity between samples. So, by adjusting the value of α can optimize the adaptive gain control module, then improve the quality of the synthesized speech.

The result shows the highest synthesized speech quality is achieved when the low-pass filter coefficient $\alpha=0.94$, meanwhile, the adaptive gain control reaches the best state.

3.3. Improved fixed codebook gain smoothing

The smoothing of the fixed codebook gain is processed according to two parameters of the speech, namely the degree of voiced and stability. The differences between the average vector and the quantized interpolation vector can reflect the stability of the speech. Therefore, the distance between the LSFs is defined as

$$diff_m=\sum_{i=0}^{10}\left[f^{[0]}(i)-f^{[-1]}(i)\right]^2 \tag{12}$$

where $f^{[0]}(i)$ is the LSFs of the current frame and $f^{[-1]}(i)$ is the LSFs of the previous frame. By smoothing the fixed codebook gain, the energy fluctuation of the excitation signal can be reduced, thereby improve the quality of the synthesized speech.

3.4. Improved anti-sparse processing

The improved anti-sparse processing is shown as below

Improved anti-sparse processing algorithm
if $\hat{g}_p < 0.4$:
$impNr = 0$;
else if $\hat{g}_p < 0$:
$impNr = 1$;
else:
if $\hat{g}_p > 0.9$:
$impNr = 3$;
else:
$impNr = 2$;

Four pre-stored impulse responses are used and a number $impNr = 0,1,2,3$ is set to select one of them. A value of 3 indicates no modification, a value of 2 indicates a slight modification, a value of 1 indicates a moderate modification, while a value of 0 indicates a great modification. After optimization, the balance between maintaining the original speech waveform and improving the perceived quality of the lower bit-rate modes is achieved.

4. Experiment results and analysis

Experiments are done with the NTT-AT Chinese speech database, which contains 96 sentences from 4 females and 4 males. The sampling rate for each speech sample is 16 kHz, and the data format is 16-bit PCM with a duration of 8s [7].

Perceptual evaluation of speech quality (PESQ) is an objective speech quality evaluation method proposed by the ITU organization in 2001 [8]. And the PESQ method is used to evaluate the synthesized speech quality. Fig.2 shows the comparison of average mean opinion score (MOS) scores for each bit-rate mode before and after AMR-NB optimization.

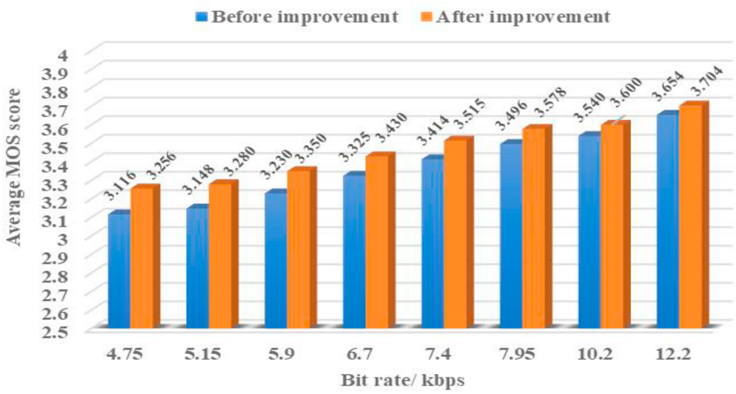

Fig. 2. Comparison of the reconstructed speech quality between the AMR-NB speech codec and the improved AMR-NB speech codec.

It can be seen that the synthesized speech quality of AMR-NB in all bit-rate modes is improved after the post-processing algorithm is optimized. Actually, except for the formant filter and the adaptive gain control modules, the anti-sparseness processing and the fixed codebook gain smoothing are not used in the 12.2kbps mode. So, the speech quality in the 12.2kbps mode is improved about 0.050 of MOS score maximally. Meanwhile, the 10.2kbps mode is not included the anti-sparseness processing module, and the 7.95kbps mode is not included the fixed codebook gain smoothing module. Besides the parameter tuning is mainly based on the 4.75kbps, 5.15kbps and 5.90kbps modes. Therefore, the improvements in the 7.95kbps and 10.2kbps modes are smaller. As a result, the lower bit-rate mode achieves the bigger improvement.

We also evaluate the complexity of the optimized post-processing algorithm on Lenovo Xiaoxin air laptop, intel i7-8565U processor, 1.80GHz, and 8G RAM. The detailed results are shown in Table.1.

Table 1. Computational complexity of AMR-NB decoder.

Bitrate	Average running time (ms)		
(Kbps)	Before improvement	After improvement	Increment
4.75	89.792	90.625	0.928%
5.15	88.750	93.646	5.516%
5.90	90.625	91.771	1.264%
6.70	90.625	93.958	3.678%
7.40	87.917	90.521	2.962%
7.95	92.188	92.916	0.790%
10.2	90.521	91.875	1.496%
12.2	90.833	93.437	2.867%

According to Table 1, the computational complexity of the improved AMR-NB decoder is slightly increased ranging from 0.928% to 5.516%. Especially in the 4.75Kbps mode which obtains the biggest MOS score improvement, the complexity increment is 0.928%. Although the 5.15kbps mode gets the largest complexity increment of 5.516%, the running time is just increased by 4.896ms for each speech sample. So, on the basis of less increase in computational complexity, an obvious improvement of the synthesized speech quality in AMR-NB has been achieved.

5. Conclusions

In this paper, we propose an improved post-processing algorithm for AMR-NB codec, by optimizing the formant filter, the adaptive gain control, the fixed codebook gain smoothing and the anti-sparse processing. The experimental results show that AMR-NB codec with the improved post-processing algorithm achieves an obvious improvement of the synthesized speech quality at the price of a slight increase of the computational complexity， especially for the lower bit-rate modes. Therefore, the improved post-processing algorithm can be adopted in AMR-NB codec for wider application.

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